

Technical Report: Predicting School ACT Performance from Socioeconomic Factors

Research Question: Is average high school ACT performance predicted by socioeconomic factors, and if so, which factors are most influential?

1. Introduction

This report addresses whether school-level ACT performance can be predicted from socioeconomic and financial variables. The ACT serves as a critical college admissions benchmark, and understanding its determinants has significant policy implications for educational equity and resource allocation. We hypothesized that socioeconomic status would significantly predict ACT scores, potentially outweighing the influence of school spending.

2. Data Integration and Preprocessing

2.1 Data Sources

Three datasets were integrated to create the analytical dataset:

1. **School Information Dataset:** School-level demographics, performance, and institutional characteristics (school_name, state, zip_code, district_id, rate_unemployment, percent_college, percent_married, median_income, percent_lunch, average_act, school_type, school_level, charter)
2. **EdGap Dataset:** Additional socio-economic and educational characteristics at the school/district level
3. **School Finance Dataset:** Financial metrics including expenditure_per_pupil and enrollment

2.2 Critical Integration Challenge: Granularity Mismatch

The primary technical obstacle was a structural incompatibility between datasets. The School Information and EdGap data operated at the **school level** (individual schools), while the Finance dataset provided data at the **district or state level** (aggregated administrative units). Direct merging was impossible due to incompatible join keys.

Solution: We created a new variable, extracted_district_id, by parsing and standardizing the existing district_id field. This harmonized key enabled a composite merge strategy using [state, extracted_district_id] as the join keys. District-level financial metrics were then propagated to all schools within each district. The successful merge produced merged_df, combining demographic, socioeconomic, performance, and financial dimensions.

2.3 Data Cleaning and Quality Control

Missing Data Treatment: Implemented state-median imputation: expenditure_per_pupil missing values were replaced with state medians

Invalid Value Handling: median_income values < \$1,000 set to NaN (implausible), then imputed with state medians. All corrected values validated to fall within realistic ranges

Filtering for Analytical Focus:

- Retained only school_level == "High" (ACT is a high school assessment), resulting in 7,227 schools
- Removed records with missing average_act (dependent variable)

Final Validation: Confirmed zero missing values across all core variables (rate_unemployment, percent_college, percent_married, median_income, average_act, percent_lunch, expenditure_per_pupil, enrollment).

Export: merged_df.to_csv('education_clean_new_2.csv', encoding='utf-8-sig', index=False)

3. Exploratory Data Analysis

3.1 Distributional Assessment: Visual and statistical diagnostics revealed severe positive skewness in enrollment, expenditure_per_pupil. To satisfy regression normality assumptions and stabilize variance, we applied logarithmic transformations on those.

3.2 Correlation Analysis: Correlation analysis identified key relationships:

- **Dominant negative correlation:** percent_lunch (free/reduced lunch eligibility, a poverty proxy) with average_act: $r \approx -0.78$
- **Strong positive intercorrelations** among socioeconomic variables:
 - percent_college $\times$ median_income: $r \approx +0.71$
 - percent_married $\times$ median_income: $r \approx +0.65$
- **Weak correlations** with ACT:
 - expenditure_per_pupil: $r \approx 0.10$
 - enrollment: $r \approx 0.05$

Key Insight: Poverty exhibits the strongest bivariate relationship with ACT performance, while school spending shows negligible association. High intercorrelations among predictors signaled potential multicollinearity issues requiring careful model specification.

3.3 Outlier Detection: Used IQR method (values beyond $1.5 \times \text{IQR}$ from Q1/Q3) to identify outliers. Outliers were retained as they represented legitimate extreme values (e.g., high-spending districts, large schools), and log transformations sufficiently reduced their leverage.

4. Regression Modeling

4.1 Simple Linear Regression: Median Income

Model: $\text{Average_ACT} \sim \beta_0 + \beta_1 \times \text{Median_Income} + \varepsilon$

Results:

- $R^2 = 0.211$ (weak explanatory power)
- Residual plot revealed non-linearity (diminishing returns at high income levels) and heteroscedasticity (funnel-shaped pattern)

Quadratic Extension: Adding Median_Income^2 improved fit slightly ($R^2 = 0.220$) but introduced extreme numerical instability (Condition Number $\approx 2.28 \times 10^{10}$) due to severe multicollinearity between income and its square.

Conclusion: Median income alone is insufficient for prediction and exhibits non-linear effects.

4.2 Full Multiple Linear Regression

Model: Model: We predicted average ACT scores using seven variables: unemployment rate, college education percentage, married household percentage, median income, poverty rate (lunch eligibility), log-transformed spending per pupil, and log-transformed enrollment.

Results:

- $R^2 = 0.629$ (strong fit)
- $\text{MAE} = 1.14$ ACT points (excellent prediction accuracy)
- **Significant predictors ($p < 0.05$):** percent_lunch, percent_college, rate_unemployment
- **Non-significant predictors:** median_income, percent_married, log_exp_per_pupil, log_enrollment

Critical Problem: Condition Number $\approx 2.5 \times 10^6$ indicated severe multicollinearity, rendering individual coefficient estimates unstable and uninterpretable.

Key Finding: Once strong socioeconomic variables were included, median income and school spending effects became statistically indistinguishable from zero, their predictive information was absorbed by poverty and educational attainment measures.

4.3 Reduced Model: Eliminating Multicollinearity: To achieve stability while retaining explanatory power, we created a reduced model using only the three significant predictors:

Results:

- $R^2 = 0.628$ (virtually identical to full model)
- $\text{MAE} = 1.145$ points
- Condition Number = 25.9 (dramatically improved from 2.5×10^6)

Conclusion: The four excluded predictors were redundant. The reduced model achieved equivalent performance with greatly improved numerical stability.

4.4 Final Normalized Model: Interpretable Coefficients

To enable direct comparison of relative predictor importance, we standardized all three predictors (mean = 0, SD = 1) using scikit learn's Standard Scaler to standardize

Final Model review:

- $R^2 = 0.628$, Adjusted $R^2 = 0.627$
- MAE = 1.145 points
- **Condition Number = 1.93** (perfect numerical stability)
- F-statistic highly significant ($p < 0.001$)

Coefficient Estimates:

Predictor	Normalized Coefficient	Impact per 1 SD Increase
percent_lunch_norm	-1.777*	-1.78 ACT points
percent_college_norm	+0.283*	+0.28 ACT points
rate_unemployment_norm	-0.123*	-0.12 ACT points

All coefficients significant at $p < 0.001$

Interpretation: The standardized coefficients reveal relative magnitudes of effect. A one-standard-deviation increase in poverty (percent_lunch) decreases ACT scores by 1.78 points, **six times larger** than the effect of educational attainment (percent_college, +0.28 points) and **fifteen times larger** than unemployment (-0.12 points).

5. Model Validation

5.1 Residual Diagnostics

- **Homoscedasticity:** Residual plot showed random scatter with relatively constant variance (no funnel pattern)
- **Normality:** Q-Q plot confirmed residuals approximately normally distributed
- **Independence:** No autocorrelation structure in cross-sectional data

5.2 Influential Observations: Cook's Distance analysis revealed no observations exceeding the $4/n$ threshold. The model is robust to individual data points.

5.3 Prediction Accuracy: MAE = 1.145 points represents 4-7% error relative to typical high school ACT score ranges (16-25). This is excellent accuracy for real-world educational data.

6. Results and Discussion

Research Question Answer: Yes, socioeconomic factors significantly predict school-level ACT performance, explaining 62.8% of variance.

Key Results:

1. **Poverty is the dominant predictor:** percent_lunch has by far the largest effect magnitude, confirming socioeconomic disadvantage as the primary driver of ACT performance differences
2. **Educational attainment matters, but less:** Community college education levels positively influence ACT scores but with substantially smaller effect than poverty
3. **School spending is not significant:** expenditure_per_pupil shows no independent association with ACT performance when controlling for socioeconomic factors
4. **Median income effect is fully mediated:** Income's influence operates entirely through poverty rates and educational attainment

6.2 Limitations

1. **District-level financial data:** Using district-aggregated spending for school-level analysis reduced within-district variation and may have attenuated expenditure effects
2. **Cross-sectional design:** Cannot establish causation; unobserved confounders may influence observed relationships
3. **Missing NY data:** Complete expenditure unavailability for New York limits geographic scope
4. **Aggregation bias:** School-level averages may obscure individual student-level variation

7. Conclusion

This analysis definitively demonstrates that socioeconomic factors, particularly poverty as measured by free/reduced lunch eligibility, powerfully predict high school ACT performance. Using a carefully integrated dataset combining school demographics, socioeconomic indicators, and financial metrics (n=7,227 high schools), we developed a stable, interpretable three-predictor model explaining 62.8% of ACT score variance with a 1.15-point average prediction error.

The normalized coefficients reveal that poverty's negative effect is six times larger than educational attainment's positive effect and that school spending has no detectable independent impact. This confirms that the educational achievement gap is fundamentally a socioeconomic gap, underscoring that educational equity requires addressing systemic inequality as a prerequisite for improved outcomes.

Technical Reproducibility: The complete analysis workflow from extracted_district_id creation for merging through log transformations, standardization, and iterative model refinement is documented herein, achievable in any statistical software with the cleaned dataset education_clean_new_2.csv.